

Environmental Studies: Impact Assessment & Solar Energy Trans

Subject: ESS - Environmental Studies & Renewable Systems | Author: UNEP / OpenStax / EduSync Sustainability E
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1. Environmental Impact Assessment (EIA) Framework

EIA is a statutory process consisting of Screening, Baseline Scoping, and Impact Prediction.
Leopold matrices evaluate magnitude and importance of developmental interventions.
Public hearing and Environmental Management Plans (EMP) ensure mitigation compliance.

2. Solar Photovoltaic (PV) & Energy Conversion

Semiconductor p-n junctions convert incident photons into electron-hole pairs via PV effect.
Shockley-Queisser theoretical efficiency limit governs single-junction silicon cells (~33%).
Decentralized microgrids with battery storage enable resilient carbon-neutral architectures.

3. Carbon Accounting & Ecosystem Biodiversity

Scope 1 (direct), Scope 2 (purchased electricity), and Scope 3 (value chain) carbon footprints.
Shannon-Wiener diversity index and Simpson index quantify ecosystem resilience.